

M. Bala Vighnesh Reddy

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EDUCATION

Indian Institute of Information Technology, Manipur

Bachelor of Technology in Computer Science and Engineering

Nov 2021 – Apr 2025

WORK EXPERIENCE

Computer Vision Intern

BigThinX, Bengaluru

Aug 2025 – Feb 2026

- Built a computer vision pipeline for the fashion domain, combining **U²-Net** and **P3M-Net** for foreground segmentation with **DWPose** and **MediaPipe** for pose estimation. Added custom heuristic rules to improve accuracy on 2-view images.
- Used the **SMPL parametric body model** to generate 3D meshes from 2D images. Improved accuracy over previous methods, particularly for **loose-fitting garments** where body outlines are difficult to detect.
- Enhanced low-resolution and low-light image quality using **LLFormer** and **KANT**, reducing errors and improving the existing pipeline's performance by **30%**. Deployed the complete system on a virtual machine using a **Gradio** interface to provide real-time inference.

AI/ML Intern

IF MedTech, Navi Mumbai

Jan 2025 – May 2025

- Fine-tuned a **Vision Transformer (ViT)** on a small **medical imaging** dataset for diagnostic classification. Quantized the model to **INT8**, achieving **89% accuracy** while significantly reducing model size and inference time.
- Built an object detection pipeline using **Faster R-CNN** to accurately locate anatomical regions. Used transfer learning and data augmentation to overcome limited training data.
- Set up a semantic segmentation workflow using **DeepLabV3** for medical images, achieving a **0.64 IoU** score on a small annotated dataset.
- Implemented a **RAG application** for medical queries over a reference book, retrieving relevant context for each question from stored **vector embeddings** and passing it to a **Llama3-8B** model for context-grounded answers.

PROJECTS / SIDE BUILDS

Text-to-SQL Research : Qwen3-4B-Instruct, DeepSeek, QLoRA, GRPO, Tinker API

- Applied knowledge distillation from DeepSeek to **Qwen3-4B** using **LoRA**. Generated over 9,000 Chain-of-Thought SQL examples to train the model on the **Spider** and **BIRD** datasets.
- Wrote execution-based reward functions for **GRPO** to evaluate SQL correctness in a reinforcement learning setup, improving performance by **10%** over the baseline.
- Achieved **86.7% pass@4** on the Spider dev set and **70.7% pass@4** on the BIRD dev set. Ran all training experiments efficiently using 4-bit QLoRA on the **NVIDIA RTX PRO 6000 Blackwell**.

AutoMCQGen (Automated MCQ Generator) : Llama2-7B, GPT-3.5, NLP, LMQL

[GitHub](#)

- Built a multiple-choice question generator using **Llama2-7B** and **LMQL**, constraining model outputs to follow strict schemas and remain aligned with the academic subject.
- Addressed the **4,096-token context window limitation** of Llama2-7B by designing an **NLP preprocessing pipeline** with rule-based text splitting and topic modeling to convert long documents into logical, topic-aware chunks.
- Set up an **LLM-as-judge** evaluation using **GPT-3.5** to check question correctness and relevance, achieving **87% content alignment** using chain-of-thought prompting.

Document Querying (RAG System) : LlamaIndex, Node.js, React, MongoDB

[GitHub](#)

- Developed a **Retrieval-Augmented Generation (RAG)** system using **LlamaIndex** to allow users to search and query large document collections accurately.
- Created **Node.js REST APIs** for uploading, indexing, and querying documents in **MongoDB**. Built a **React.js** frontend with expandable chat threads.
- Implemented secure user authentication and session management to protect data privacy.

TECH SKILLS / FRAMEWORKS

- **Languages & Frameworks:** Python, C/C++, JavaScript, HTML, CSS, FastAPI, Flask, GitHub, Ollama, HuggingFace, OpenRouter
- **Web Development:** MERN Stack, REST APIs, Express.js, Node.js, React.js, Tailwind CSS, Bootstrap
- **AI & Machine Learning:** PyTorch, HuggingFace, Transformers, LLMs, Finetuned Models, RAGs, VectorDB, LangChain, LlamaIndex, UnSloth, CrewAI, MCP, Ollama
- **Databases & Cloud:** MongoDB, MySQL, GCP

CERTIFICATIONS

- **Generative AI with Large Language Models** – DeepLearning.AI (2024)
- **Neural Networks and Deep Learning** – DeepLearning.AI / Coursera (2024)